BRFSS 2021 Health Status Classification

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Abstract

We compare five classifiers for identifying respondents who report fair or poor health in BRFSS 2021. After target cleaning, the dataset contains 437,532 respondents, including 72,741 positive cases (16.63%), and 108 original predictors. We evaluate the full feature set, SelectKBest, TruncatedSVD, RFECV, four engineered features, class weighting, and nested hyperparameter optimization. CatBoost gives the strongest ranking performance: ROC-AUC is 0.9023 and average precision is 0.6986 in three-fold nested cross-validation. With class weighting, its recall reaches 0.8051 at precision 0.4934. In the subsequent paired ten-fold comparison, CatBoost exceeds logistic regression in every fold, with a mean ROC-AUC difference of 0.00323 and Wilcoxon $p = 0.00195$. SHAP highlights physical health, functional limitations and chronic disease burden. Feature engineering adds virtually no discrimination; RFECV offers a compact alternative. The results describe concurrent self-reported health, with limitations from survey design, subjective labels and reuse of the dataset during model selection.

1 Introduction and problem definition

Self-rated general health combines physical, mental, functional and socioeconomic aspects of a respondent's situation. This project develops and compares binary classifiers for predicting whether a respondent reports fair or poor health, using medical, behavioral and demographic variables from the same survey.

The target is derived from `GENHLTH`: excellent, very good and good health form class 0, while fair and poor health form class 1. We ask which classifier and feature representation give the best discrimination, whether a smaller representation preserves performance, and how class weighting changes detection of the minority class. We also test whether clinically motivated summaries improve prediction, compare the two selected models statistically, and examine their errors and explanations.

This is a cross-sectional classification task. The models reconstruct a respondent's current health rating from concurrent questionnaire responses; they do not estimate future disease incidence or establish causal effects. ROC-AUC is the primary model-selection criterion. Average precision, F1, precision, recall and balanced accuracy characterize performance under class imbalance.

Project materials. Source code and video presentation: `https://github.com/tomabes/health_survey_ml`.

2 Dataset and supporting literature

BRFSS is a CDC-supported telephone survey of noninstitutionalized adults aged 18 years or older. It collects self-reported information about risk behaviors, chronic conditions, healthcare access and preventive services. The 2021 combined file includes 49 states, the District of Columbia, Guam, Puerto Rico and the US Virgin Islands (CDC, 2022a, 2023). The `LLCP2021.csv` has **438,693 rows and 303 columns**.

The official dataset landing page provides downloads and documentation. The codebook defines variables and response codes; the calculated-variable guide explains CDC-derived indicators (CDC, 2022b,c). The overview and weighting guide describe sampling and appropriate population analyses (CDC, 2022a,d).

Clark et al. (2021) used BRFSS 2017, with replication in 2016, to predict self-rated health using regularized logistic regression, random forest and SVM. Their reported AUC values were approximately 0.80–0.81, and socioeconomic variables were prominent predictors in mid-life. Their positive outcome was excellent or very good health, so these AUC values are contextual evidence, not directly comparable benchmarks for our fair-or-poor target. The systematic review by Pierannunzi et al. (2013) found that BRFSS estimates were generally comparable with other self-report surveys, while agreement with physical measurements was less consistent.

Figure 1 displays the five-category and binary target distributions among respondents with valid health ratings; exact counts are given in Table 18 in the appendix. The positive-to-negative imbalance is approximately 1:5.

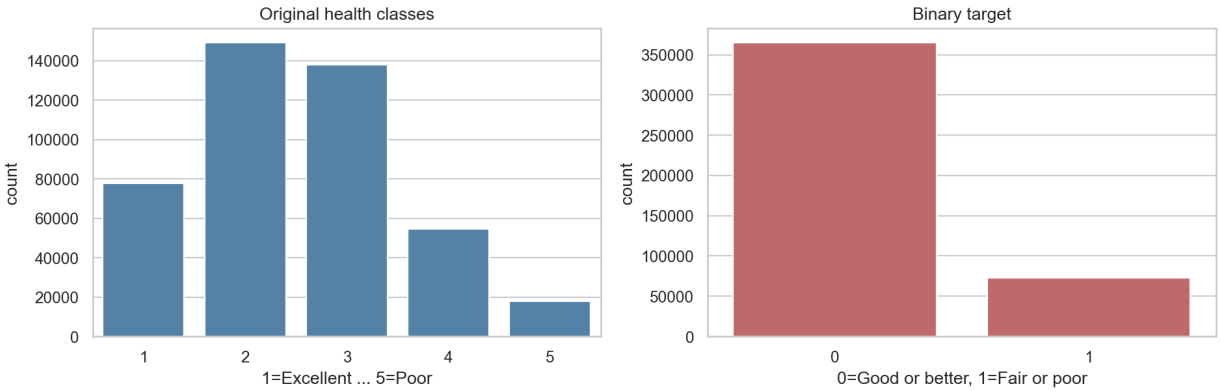


Figure 1: Original five-category health ratings and the binary target after target cleaning.

3 Data loading, audit and cleaning

The audit found **zero fully duplicated rows** and **zero duplicated respondent identifiers** using `_STATE` and `SEQNO`. Table 1 summarizes the successive dimensions. We remove variables related to the mechanics of the telephone survey while retaining medical, behavioral and demographic predictors. These 51 exclusions cover record metadata (state, interview dates, disposition, sequence number, primary sampling unit), landline and cell-phone routing questions, telephone ownership, questionnaire version and language, and survey-design and weighting variables (strata, raking factors and final weights). None of them describes the respondent's health, and the design variables would allow models to learn sampling artefacts.

Table 1: Data preparation and dimensionality before encoding.

Stage	Rows	Columns
Original CSV	438,693	303
Remove 51 service variables and `_RFHLTH`	438,693	251
Keep valid target and add binary target	437,532	252
Remove 3 constant or empty variables	437,532	249
Remove 129 variables above 50% blank	437,532	120
Remove 9 raw variables with numeric analogs	437,532	111
Remove duplicate height representation	437,532	110
Exclude both target fields from predictors	437,532	108

Target leakage. The CDC-calculated variable `_RFHLTH` ("adults with good or better health") is computed directly from `GENHLTH`: ratings 1–3 are coded 1, ratings 4–5 are coded 2, and uncertain or refused answers are coded 9 (CDC, 2022c). It is therefore a relabeled copy of our binary target and is removed. No other variable in the file is derived from `GENHLTH`. Unlike Clark et al. (2021), we deliberately retain physically, mentally and activity-limited unhealthy days (`PHYSHLTH`, `MENTHLTH`, `POORHLTH`) and their CDC categorizations `_PHYS14D` and `_MENT14D`. They are separate questions with a fixed 30-day reference period and are part of the health profile. They are nevertheless conceptually close to self-rated health, so the high discrimination reported below partly reflects this construct overlap (Section 19).

The 788 "do not know", 369 refused and 4 missing `GENHLTH` responses cannot define a health class; excluding these **1,161 rows** leaves **437,532 respondents**. Three constant or entirely empty variables, `TOLDCFS`, `HAVECFS` and `WORKCFS`, are removed. Variable names are then replaced with readable labels (for example, `PHYSHLTH` becomes `physical_days` and `_PHYS14D` becomes `_physical_status`); a leading underscore marks CDC-calculated variables.

Features with more than 50% missing values are removed. Most belong to optional survey modules or contain detailed follow-up questions asked only to specific subgroups, such as respondents with cancer, diabetes, asthma or arthritis. The threshold corresponds to a clear gap in the observed missingness distribution: the last removed feature has 51.73% missing values, while the next retained feature has 46.85%. General diagnostic indicators for diabetes, asthma, cancer, cardiovascular disease, COPD, kidney disease and arthritis are retained and have less than approximately 1% blank values. This filtering removes poorly covered conditional details while preserving the main medical, behavioral and demographic information.

Several CDC-calculated variables recode raw answers that are also retained, for example `_arthritis` and `arthritis`, `_sex` and `sex`, four age representations and five race variables. This redundancy is kept because tree ensembles and regularized linear models tolerate it, and the recodes sometimes merge uncertain answers in a useful way; its cost is that importance is shared between correlated columns in the SHAP analysis.

4 Missing values and predictor representation

The initial retained predictors comprise 13 numerical, 15 mixed-code and 90 categorical variables. Nine raw variables are dropped because CDC provides decoded equivalents: encoded weight and height (replaced by weight in kilograms and height in meters), alcohol frequency (replaced by drink occasions per day) and six food-frequency questions (replaced by daily frequencies). The remaining mixed-code variables—physical, mental and activity-limited days, children, daily drinks and weekly drinks—are treated as categorical. For example, 88 denotes no unhealthy days, while 77 and 99 denote uncertainty and refusal; their numeric codes must not be interpreted as ordinary counts (CDC, 2022b).

EDA is performed on the cleaned variables before one-hot encoding. Blank rates are computed on the unimputed data; for the EDA plots, categorical blanks are coded -1 and numerical blanks are median-imputed. Figure 2 shows the 30 largest remaining blank rates: activity-limited days 46.85%, transgender status 43.99%, BMI 10.64%, weight 9.22% and height in meters 5.57%.

These blanks are often structural: activity-limited days is asked only after at least one unhealthy day, so a blank there means "none" and the -1 category is informative.

For modeling, imputation is fitted independently within each training fold. Numerical predictors use median imputation with missingness indicators and standardization. Categorical predictors use constant -1 imputation and sparse one-hot encoding with unseen validation categories ignored. Survey codes already present in the data remain explicit categories. After removing the duplicate height-in-inches representation, the final matrix contains **12 numerical and 96 categorical**

predictors.

No outliers are removed. Implausible food-frequency values (up to 99 times per day) are kept, as CDC already flags them in `_FRT16A` and `_VEG23A`, and they affect only the standardized inputs of the linear models.

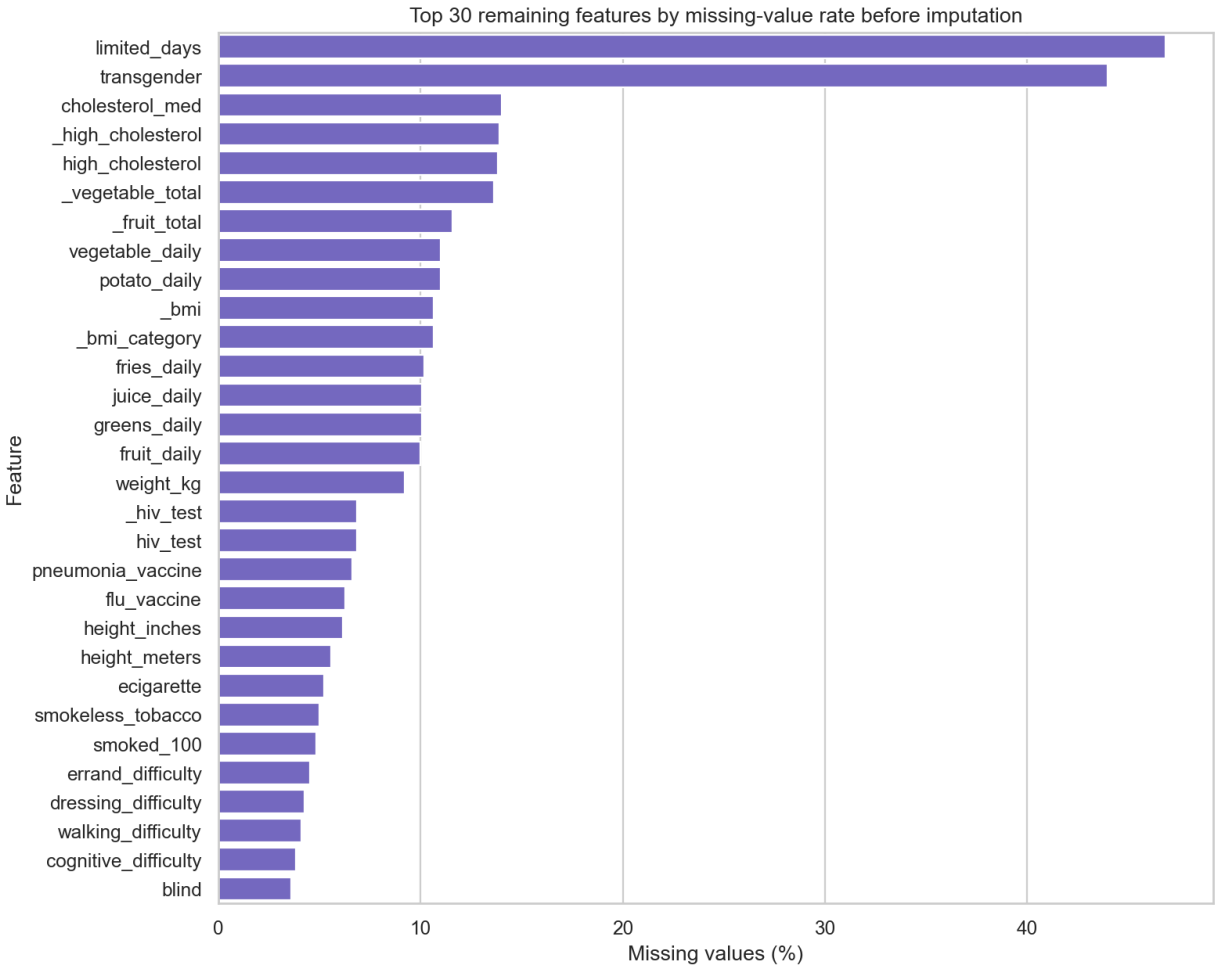


Figure 2: Top 30 retained variables by blank-value rate before imputation. Special response codes are separate from blanks.

5 Exploratory data analysis

5.1 Age, BMI and reported diagnoses

Figure 3 shows that respondents in the fair-or-poor group tend to be older and have higher BMI. Figure 4 shows higher reported diagnosis frequencies in that group: cancer is 25.27% versus 15.11%, asthma 23.44% versus 12.14%, and diabetes 30.42% versus 9.67%. Diabetes is therefore approximately 3.15 times as frequent in the positive class.

Cancer means skin cancer or another cancer diagnosis; asthma means ever diagnosed; diabetes excludes gestational diabetes and prediabetes. The indicators count explicit affirmative answers, so the complementary group also includes uncertain, refused or blank responses rather than only confirmed negative diagnoses. Figure 15 in the appendix further compares age and BMI by these indicators: cancer and diabetes are associated with older age profiles, while asthma and diabetes show higher BMI profiles.

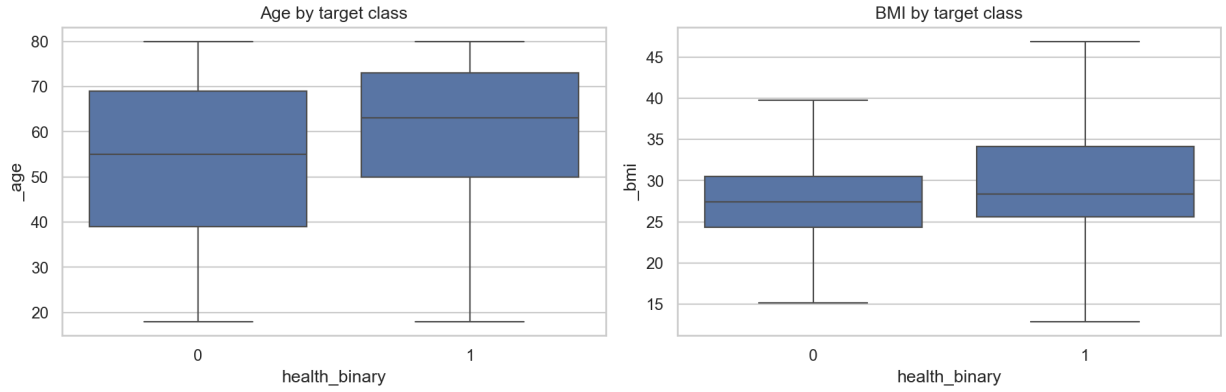


Figure 3: Age and BMI by target class. Class 0 is good or better; class 1 is fair or poor.

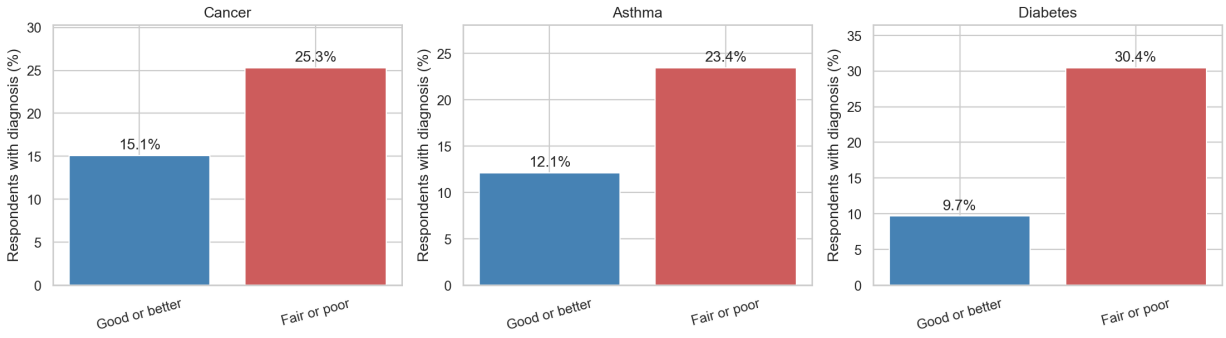


Figure 4: Percentages reporting cancer, asthma or diagnosed diabetes within each target class.

5.2 Behavioral, socioeconomic and clinical associations

Figure 5 shows the share of each target class within the categories of 12 selected variables; category sizes are given under each bar. Sex shows little marginal association with the target. Without physical activity, the proportion reporting fair or poor health is approximately three times higher. Daily smokers report fair or poor health more frequently than never-smokers. Income and education show pronounced gradients. Heart attack, coronary heart disease, stroke, diabetes, COPD and kidney disease are strongly associated with fair or poor health. Depression also shows a substantial association.

5.3 Numerical correlations and redundancy

Spearman correlations with the binary target are weak (Figure 6). The largest are age (0.13), BMI (0.12), leafy-green consumption (-0.11), weight (0.08), and total vegetable consumption (-0.08). A positive coefficient indicates association with fair or poor health; a negative coefficient indicates association with good or better health. A weak individual correlation does not imply that a feature is useless: relationships may be nonlinear or emerge jointly with other variables.

Height in inches and height in meters have correlation 1.00, so only height in meters is retained. Other strong correlations are fruit frequency with total fruit consumption (0.85), vegetable frequency with total vegetable consumption (0.75), and leafy greens with total vegetable consumption (0.62). Weight and BMI correlate at 0.83, but are not exact duplicates because BMI also incorporates height; both are retained.

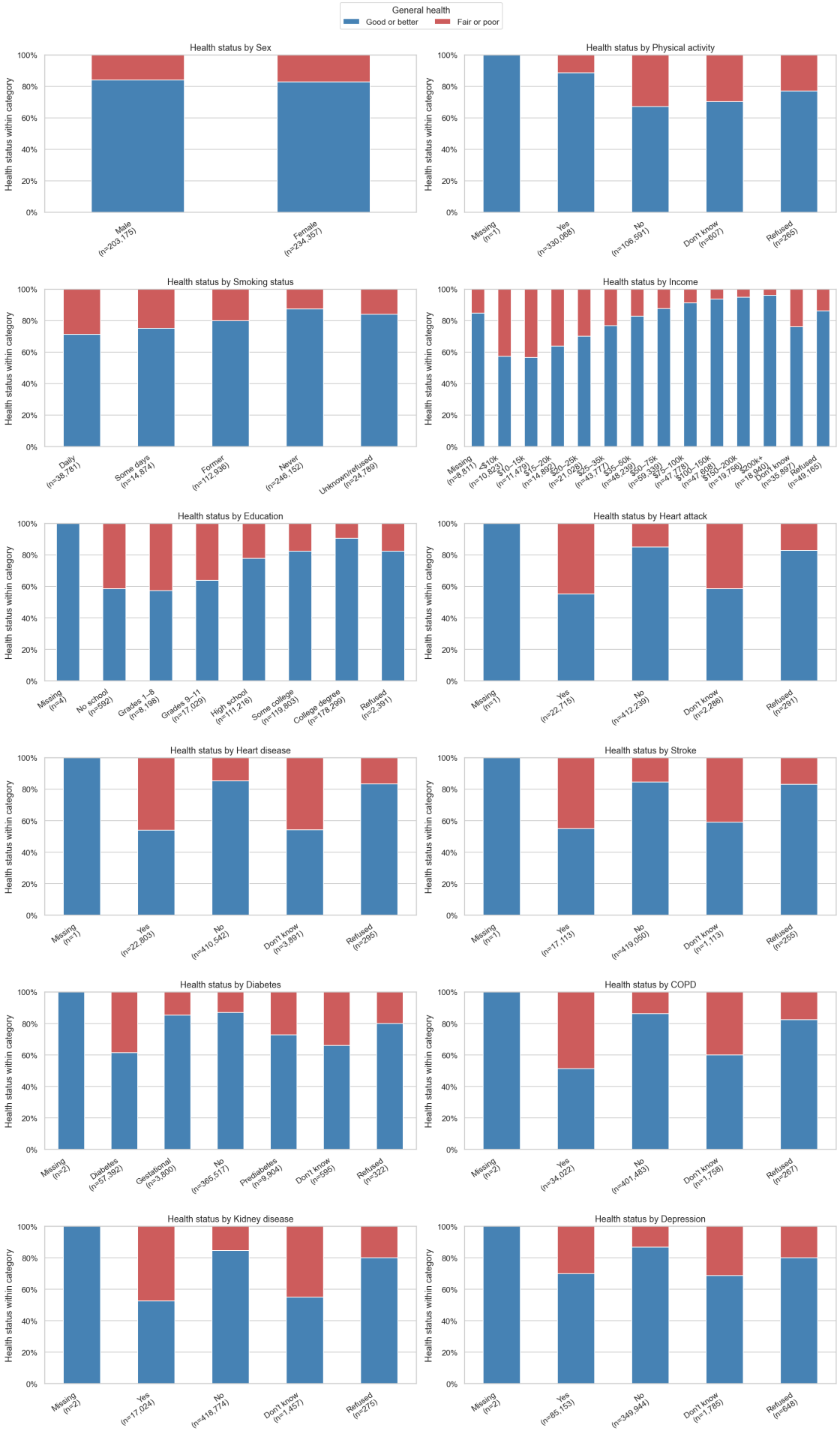


Figure 5: Health status within several categories. Blue is good or better health; red is fair or poor health.

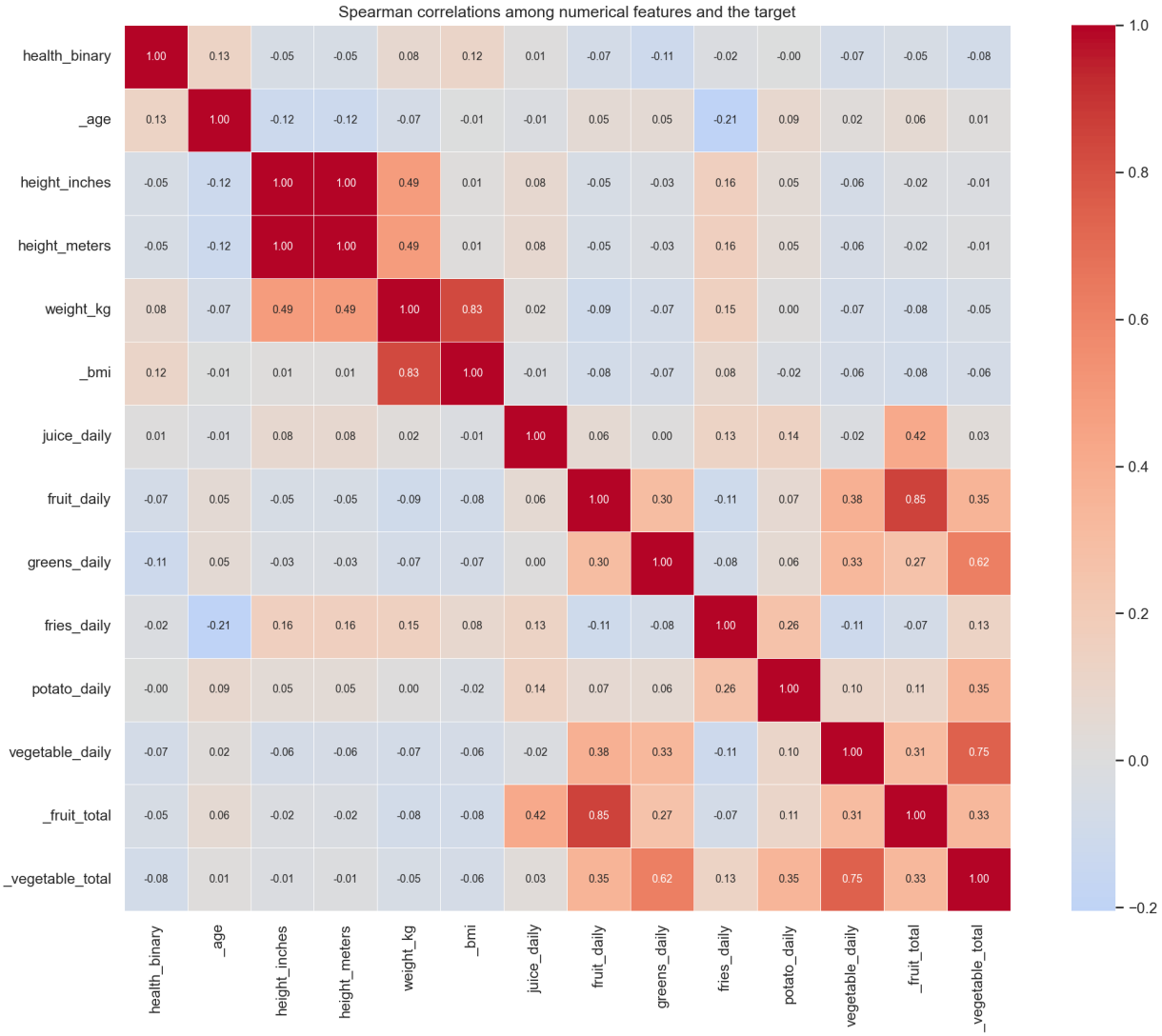


Figure 6: Spearman correlations among numerical variables and the target before removal of height in inches.

6 Preprocessing, classifiers, validation and metrics

The predictor matrix X has shape $437{,}532 \times 108$ and excludes both the original health rating and its binary derivative. The target vector y contains 364,791 negatives and 72,741 positives. A common random seed of 42 and shared stratified splits support paired comparisons.

The implementation uses pandas and NumPy for data handling, scikit-learn for preprocessing, selection and evaluation, CatBoost for boosting, SciPy for the Wilcoxon test, and SHAP for explanations. Matplotlib and seaborn generate the figures. Table 2 records the initial classifier settings. All classifiers receive the same encoded feature representation; CatBoost does not use a separate native categorical-data pipeline here.

Table 2: Initial classifier configurations before class weighting and tuning.

Model	Main settings
Logistic regression	$C = 1$, saga, 1,000 iterations, tolerance 10^{-3}
Decision tree	Maximum depth 15, minimum leaf size 20
Random forest	100 trees, minimum leaf size 5, square-root feature sampling
LinearSVC	$C = 1$, automatic primal or dual selection
CatBoost	300 iterations, depth 6, learning rate 0.1, Logloss

Initial settings and manual tuning. The initial settings in Table 2 were chosen manually as common starting values close to library defaults, adjusted for a dataset of 437,532 respondents. The single decision tree is limited to depth 15 with at least 20 respondents per leaf to avoid memorizing individual records; the random forest uses 100 trees and a minimum leaf size of 5 to keep training time manageable; logistic regression uses the `saga` solver, suited to large sparse inputs, with a relaxed tolerance for convergence; CatBoost uses a moderate 300 iterations at depth 6. A few alternative values were tried during development. Because a systematic, nested search was planned for the best models (Section 15), no separate manual-tuning log was kept.

Validation scheme and the 10-fold requirement. The course guidelines prescribe stratified ten-fold cross-validation for all experiments. With about 900 sparse encoded features, five classifiers, five feature representations and RFECV's own internal cross-validation, ten-fold screening would have taken roughly an order of magnitude longer than two-fold screening. Exploratory model and feature comparisons therefore use stratified **two-fold cross-validation**. Each validation fold contains about 219,000 respondents, and the standard deviation of ROC-AUC between folds is about 0.001, so the ranking of models is stable at this scale. The decisive comparison between the two final models uses **stratified ten-fold cross-validation** with identical splits, as required for the Wilcoxon test (Section 16).

Preprocessing and supervised feature selection are fitted only on each training fold, then applied to its validation fold. Nested grid search uses **three outer folds and two inner folds**; a subsequent statistical comparison evaluates fixed selected configurations on **ten identical outer splits**.

ROC-AUC and average precision use probability estimates or, for LinearSVC, decision-function scores. **Average precision (AP)** summarizes the precision–recall curve as the mean of precision values weighted by the increase in recall at each threshold. It is closely related to the area under the precision–recall curve but does not use trapezoidal interpolation, and for an uninformative classifier it equals the positive-class prevalence (0.166). **F1, precision, recall** and **balanced accuracy (BA)** use the classifiers' default decision rule. Tables report means over the validation folds. With TP, FP, TN and FN denoting confusion counts,

$$\text{Precision} = \frac{TP}{TP+FP}, \qquad \text{Recall} = \frac{TP}{TP+FN},$$

$$\text{F1} = \frac{2TP}{2TP+FP+FN}, \qquad \text{BA} = \frac{1}{2}\left(\frac{TP}{TP+FN} + \frac{TN}{TN+FP}\right).$$

7 Dummy baseline

The prior-based Dummy classifier does not distinguish classes: ROC-AUC is 0.5 and AP is 0.1663, approximately the positive-class proportion. It always predicts the more frequent good-or-better class, giving F1, precision and recall of zero and balanced accuracy of 0.5 (Table 3). Every useful model should substantially exceed this baseline. Ordinary accuracy alone would be misleading because always predicting class 0 already gives 83.37% accuracy.

Table 3: Dummy baseline, mean over two folds.

Model	ROC-AUC	AP	F1	Precision	Recall	BA
Dummy	0.5000	0.1663	0.0000	0.0000	0.0000	0.5000

8 Experiment 1: Full feature set

We use all meaningful features after cleaning and missing-value handling. The full representation has an average of **877.5 encoded features across the two training folds** (893 and 862), compared

with 937 when fitted on the complete dataset. This difference arises because the categories observed during fitting differ by training set.

CatBoost achieves the best overall discrimination on the full feature set: ROC-AUC 0.9018, AP 0.6985, F1 0.6110, recall 0.5246 and balanced accuracy 0.7431. It also has the shortest training time, about 11 seconds per fold, although fit times are not strictly comparable because CatBoost uses four threads, random forest two, and the other models one. Table 4 reports all model results.

Logistic regression and LinearSVC achieve similar ROC-AUC values of 0.8984 and 0.8981, while random forest reaches 0.8970. Random forest produces the highest precision (0.7595) but lower recall (0.4590), meaning that it predicts fair or poor health less frequently but is more often correct when doing so. Decision tree performs substantially worse, with ROC-AUC 0.8612.

Table 4: Full feature set, mean performance over two folds.

Model	ROC-AUC	AP	F1	Precision	Recall	BA	Fit s
CatBoost	0.9018	0.6985	0.6110	0.7313	0.5246	0.7431	10.7
Logistic regression	0.8984	0.6901	0.6030	0.7348	0.5113	0.7373	63.7
LinearSVC	0.8981	0.6916	0.5924	0.7441	0.4921	0.7292	69.8
Random forest	0.8970	0.6868	0.5722	0.7595	0.4590	0.7150	177.2
Decision tree	0.8612	0.6095	0.5536	0.6665	0.4735	0.7131	31.3

9 Experiment 2: Basic feature selection with SelectKBest

Hypothesis. A compact subset dominated by age, physical and mental health, chronic conditions, functional limitations and socioeconomic variables will preserve most predictive performance. Dietary and specialized survey variables are expected to contribute less.

SelectKBest uses the univariate ANOVA F statistic to retain 100 encoded features. It reduces the representation from approximately 878 features to 100. Total experiment time decreases from 12.1 to 5.4 minutes, approximately a 2.2-fold reduction, although the strongest models lose some discrimination (Table 5).

Table 5: SelectKBest with 100 encoded features, mean over two folds.

Model	ROC-AUC	AP	F1	Precision	Recall	BA	Fit s
CatBoost	0.8956	0.6880	0.5997	0.7308	0.5085	0.7356	5.0
Logistic regression	0.8908	0.6793	0.5870	0.7340	0.4891	0.7269	9.2
LinearSVC	0.8915	0.6809	0.5760	0.7485	0.4681	0.7184	3.0
Random forest	0.8910	0.6788	0.5856	0.7347	0.4869	0.7259	129.0
Decision tree	0.8686	0.6232	0.5628	0.6789	0.4806	0.7176	8.4

For CatBoost, ROC-AUC decreases from 0.9018 to 0.8956 while classifier training time falls from 10.7 to 5.0 seconds. Logistic regression and LinearSVC also lose ROC-AUC while becoming substantially faster. This indicates that removed features contain useful information, although much of the representation is redundant. Random forest loses ROC-AUC but improves F1, recall and balanced accuracy, giving a better balance between the target classes. Decision tree improves across the reported predictive metrics, suggesting that this simpler model benefits from removing weak or noisy variables.

The selected features mainly represent age and BMI; physically and mentally unhealthy days; limitations in daily activities; physical activity; cardiovascular and chronic diseases; functional limitations; education, income and employment; smoking and alcohol use; and access to healthcare. No fruit, vegetable or other dietary variable is selected. The selected `limited_days_-1.0` indicator corresponds to the skip pattern described in Section 4: it marks respondents with no unhealthy

days, which explains its strong univariate association with the target. Many selected columns are redundant, such as the yes and no indicators of the same question or a raw answer and its CDC recode, so the 100 columns cover considerably fewer distinct questions; a univariate filter cannot detect this redundancy.

Verdict. The hypothesis is supported. About 11% of the encoded representation retains most of the discrimination (CatBoost ROC-AUC −0.006, AP −0.011) while more than halving the total experiment time, and the selected variables belong to the hypothesized groups, with dietary variables absent.

10 Experiment 3: Dimensionality reduction with TruncatedSVD

TruncatedSVD is used instead of conventional PCA because one-hot encoding produces a sparse matrix. With 100 components it substantially reduces the representation but gives lower discrimination than the full feature set (Table 6). CatBoost reaches ROC-AUC 0.8940, compared with 0.8956 after SelectKBest and 0.9018 with all features. LinearSVC performs best within the SVD experiment, reaching 0.8958. The dense lower-dimensional representation is therefore more suitable for the linear models than for the tree-based models in these results.

Table 6: TruncatedSVD with 100 components, mean over two folds.

Model	ROC-AUC	AP	F1	Precision	Recall	BA	Fit s
CatBoost	0.8940	0.6814	0.5912	0.7285	0.4974	0.7302	8.6
Logistic regression	0.8953	0.6838	0.5909	0.7330	0.4949	0.7295	1.9
LinearSVC	0.8958	0.6852	0.5821	0.7448	0.4777	0.7225	4.2
Random forest	0.8843	0.6592	0.5172	0.7667	0.3902	0.6833	177.3
Decision tree	0.7991	0.5072	0.4921	0.5736	0.4309	0.6835	30.3

The explained-variance curve increases smoothly without a clear elbow (Figure 7). Approximately 50 components explain 80% of the variance and 100 explain about 91%. However, preserving variance does not necessarily preserve information most useful for classifying the target.

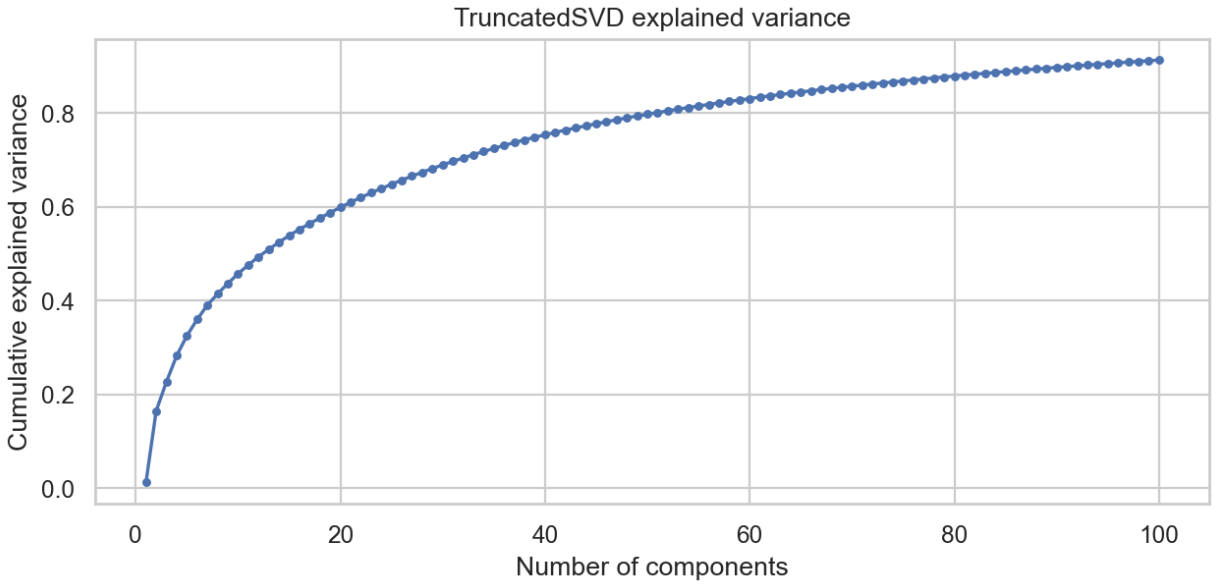


Figure 7: Cumulative explained variance of the 100 SVD components from a descriptive full-data fit.

Table 7 lists the largest absolute loadings of the first three components from a descriptive full-data fit. **Component 1** reflects a common, relatively healthy response profile: plausible fruit and vegetable

intake flags, no stroke, kidney disease, heart attack, coronary heart disease, COPD or functional limitations, together with insurance and no financial barrier to care; all its top loadings have similar magnitude (0.12–0.13).

Table 7: Six largest absolute loadings of the first three SVD components (full-data descriptive fit). "Missing" denotes the missingness indicator added by the imputer.

Component 1		Component 2		Component 3	
Feature	Loading	Feature	Loading	Feature	Loading
`_fruit_excess_1`	0.131	Missing `fries_daily`	0.295	`_age`	0.440
`_vegetable_excess_1`	0.131	Missing `greens_daily`	0.295	`_age_group_6`	0.197
`stroke_2`	0.126	Missing `juice_daily`	0.292	`_age65_group_2`	0.195
`kidney_disease_2`	0.126	Missing `potato_daily`	0.292	`_age65_group_1`	−0.193
`heart_attack_2`	0.125	Missing `vegetable_daily`	0.291	`_insured_under65_9`	0.189
`heart_disease_2`	0.124	Missing `fruit_daily`	0.289	`_insured_under65_1`	−0.165

Suffixes are survey codes: 2 is "no" for diagnosis questions; `_fruit_excess_1` and `_vegetable_excess_1` mark intakes within CDC's plausible range (at most 16 and 23 times per day); `_age_group_6` is age 65 or older; `_age65_group_1_2` is age 18–64 / 65 or older; `_insured_under65_9` marks respondents aged 65 or older, to whom the question does not apply.

Component 2 largely describes missingness in dietary variables, BMI, weight and height, alongside CDC missing or excluded-response indicators. SVD thus detects questionnaire completeness rather than a clinical condition. **Component 3** combines age and age-related conditions: age groups, arthritis, activity and work limitations, insurance, employment and pneumonia vaccination. Positive and negative loadings define opposing sides of this age and functional profile; component sign itself is arbitrary.

11 Experiment 4: Advanced feature selection with RFECV

Hypothesis. A multivariate, model-based elimination retains variables that are informative jointly with others and discards redundant indicators, which the univariate filter of Experiment 2 cannot do. A subset of roughly 150–200 encoded features chosen this way should therefore approach the full-feature ROC-AUC much more closely than the 100 SelectKBest features, at a fraction of the full dimensionality.

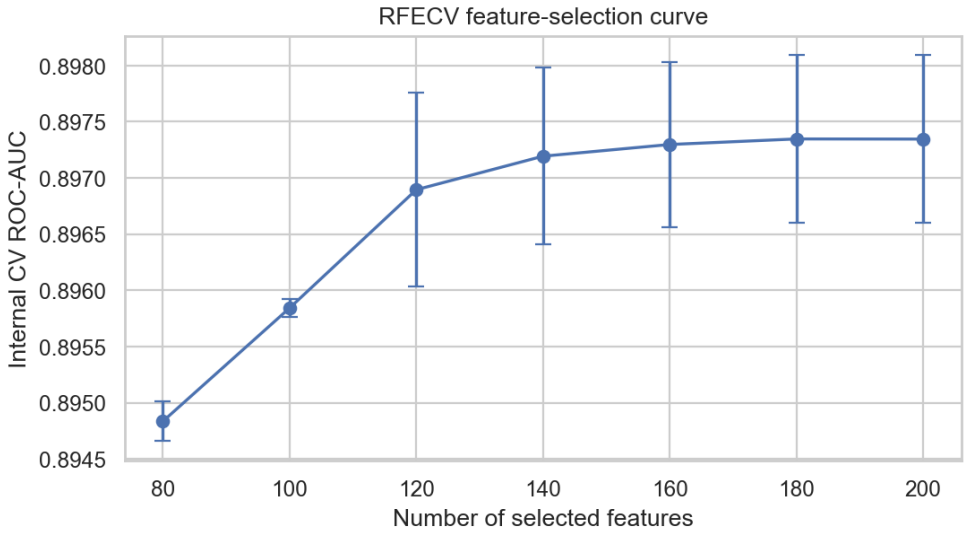


Figure 8: Internal two-fold ROC-AUC of the logistic-regression RFECV selector fitted on the full dataset. Error bars show internal-fold standard deviations.

RFECV follows an initial SelectKBest reduction to 200 features. It uses logistic regression, internal stratified two-fold cross-validation, ROC-AUC scoring, a step of 20 and a minimum of 80 retained features. The candidate sizes are therefore 80, 100, 120, 140, 160, 180 and 200.

On the full dataset RFECV selects 180 features; across the two outer training folds it retains 200 and 180, or **190** on average. The full-data selection curve (Figure 8) is almost flat between 180 and 200 features: internal mean ROC-AUC is 0.897347 and 0.897345, respectively. For CatBoost, outer ROC-AUC increases from 0.8956 with 100 SelectKBest features to 0.9004 with RFECV, approaching the full-feature result of 0.9018 (Table 8).

Table 8: RFECV representation, mean over two outer folds.

Model	ROC-AUC	AP	F1	Precision	Recall	BA	Fit s
CatBoost	0.9004	0.6953	0.6086	0.7283	0.5226	0.7419	6.1
Logistic regression	0.8973	0.6881	0.5990	0.7346	0.5056	0.7346	17.5
LinearSVC	0.8976	0.6901	0.5893	0.7467	0.4868	0.7269	6.2
Random forest	0.8959	0.6855	0.5864	0.7423	0.4847	0.7256	165.0
Decision tree	0.8657	0.6198	0.5637	0.6715	0.4858	0.7192	15.0

Verdict. The hypothesis is largely supported: RFECV recovers about 80% of the AUC lost by SelectKBest with about a fifth of the full encoded dimensionality. However, the selection curve is nearly flat from 140 features onward, and the elimination removed only 0–20 of the 200 prefiltered features. Most of the improvement therefore comes from allowing more features than in Experiment 2; here RFECV acts mainly as a principled choice of subset size rather than as an aggressive filter.

12 Experiment 5: Clinically motivated feature engineering

Hypothesis. Summary measures of multimorbidity, functional limitations and lifestyle risks, together with an age-by-comorbidity interaction, improve prediction by making cumulative burden explicit.

Four numerical features are added to the original predictors. `comorbidity_count` counts affirmative responses for 11 conditions: heart attack, heart disease, stroke, asthma, skin cancer, other cancer, COPD, depression, kidney disease, diabetes and arthritis.

`functional_limitations_count` counts six affirmative limitations: hearing, vision, cognition, walking, dressing and errands.

`lifestyle_risk_count` sums six indicators: inactivity, current smoking, overweight or obesity, binge drinking, low fruit intake and low vegetable intake. Finally,

$$\texttt{age_x_comorbidity} = \texttt{_age} \times \texttt{comorbidity_count}.$$

These counts reflect confirmed affirmative questionnaire codes, so unknown or missing answers do not add to them; they are not validated clinical severity scores. The age variable is CDC's age measure capped at 80, so the interaction inherits that cap.

Table 9: Four engineered predictors added to the full representation, mean over two folds.

Model	ROC-AUC	AP	F1	Precision	Recall	BA	Fit s
CatBoost	0.9019	0.6988	0.6110	0.7320	0.5244	0.7430	11.2
Logistic regression	0.8984	0.6901	0.6031	0.7344	0.5117	0.7374	64.8
LinearSVC	0.8982	0.6916	0.5926	0.7440	0.4924	0.7293	64.2
Random forest	0.8968	0.6862	0.5742	0.7558	0.4629	0.7166	180.1
Decision tree	0.8545	0.6065	0.5543	0.6592	0.4782	0.7145	36.1

Table 9 shows no substantial improvement. CatBoost reaches ROC-AUC 0.901859 versus 0.901803 with the full representation, a difference of only 0.000056. These new features are built from variables already present: CatBoost can find age–disease interactions, trees can form joint diagnosis rules, and counts are a compact but redundant summary of the original predictors. The hypothesis of a material predictive gain is therefore not supported. The difference of 0.000056 is more than ten times smaller than the standard deviation of CatBoost's ROC-AUC between folds (0.0008–0.0010), so the two representations are equivalent in discrimination. The single decision tree even loses ROC-AUC (0.8612 to 0.8545), possibly because the high-gain splits on the counts lead greedy splitting to partitions that generalize less well.

The engineered variant is nevertheless retained for the subsequent experiments. Since it costs nothing in discrimination and only four additional dimensions, the decision rests on interpretability: the counts give the SHAP analysis compact, clinically readable summaries of disease and limitation burden instead of importance spread over many binary indicators.

13 Comparison of feature experiments

CatBoost remains the strongest model for the full, engineered, RFECV and SelectKBest representations. The engineered variant has the highest mean ROC-AUC, but its difference from the full representation cannot be interpreted as a meaningful improvement. Figure 9 compares discrimination across every model and representation.

RFECV provides the best selection compromise for CatBoost: ROC-AUC is 0.9004 with about 190 selected features during outer validation, versus approximately 878 full encoded features. A separate full-data fit retains 180. SelectKBest is faster and more compact but reduces CatBoost ROC-AUC to 0.8956. TruncatedSVD is especially suitable for the linear models relative to the other models within the SVD experiment, but does not improve their full-feature results.

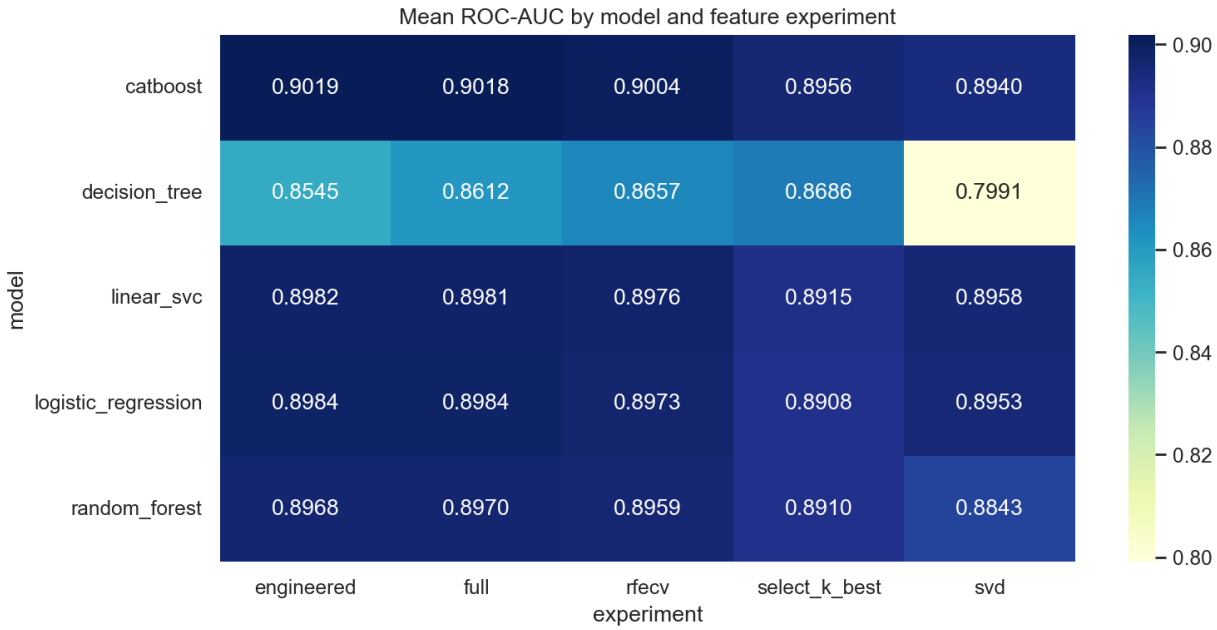


Figure 9: Mean ROC-AUC across five classifiers and five feature representations.

13.1 Training cost and total runtime

Figure 10 shows the mean classifier fit time per fold, excluding transformation and selection. Table 10 reports the total experiment runtime, which includes preprocessing, feature selection, fitting and

evaluation of all five models on both folds. RFECV reduces classifier fit time but adds selection cost, so its total 10.59 minutes remains above SelectKBest's 5.42 minutes. Fit times are wall-clock measurements on a single machine with different thread counts per model, so they indicate relative cost.

Table 10: Representation sizes and total feature-experiment runtime.

Representation	Mean output dimensions	Total minutes
Full	877.5	12.08
KBest	100.0	5.42
SVD	100.0	8.04
RFECV	190.0	10.59
Engineered	881.5	12.21

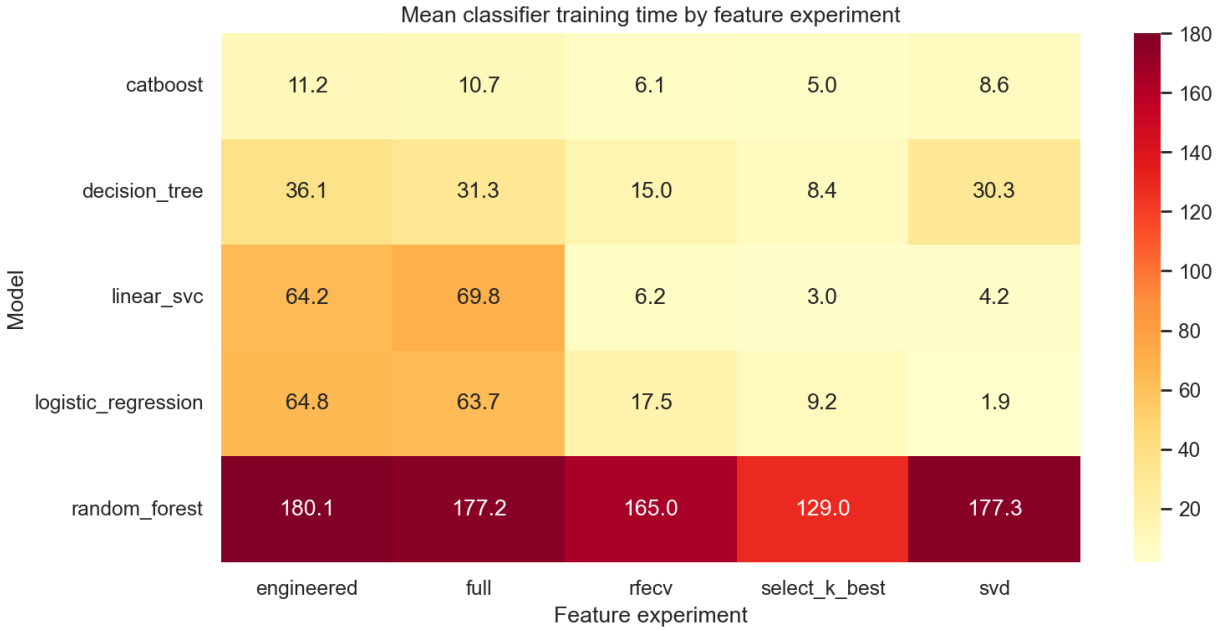


Figure 10: Mean classifier fit time in seconds for each feature experiment.

14 Class imbalance experiment

All five models are compared with and without class weighting on the engineered representation. Scikit-learn classifiers use `class_weight=balanced`; CatBoost uses `auto_class_weights=Balanced`. Weighting changes the training loss without resampling the observations.

Tables 11 and 12 show the tradeoff. CatBoost recall increases from 0.5244 to 0.8029 and balanced accuracy from 0.7430 to 0.8195, while precision falls from 0.7320 to 0.4941. Its ROC-AUC remains almost unchanged, 0.9019 versus 0.9020. This is expected: class weighting mainly shifts all scores upward, which acts like a lower decision threshold. Threshold-based metrics therefore change, while ROC-AUC and AP, which depend only on the ranking of scores, do not.

Random forest achieves the highest weighted F1 (0.6340) and precision (0.5892), but recall is lower (0.6861) than CatBoost's. Logistic regression gives F1 0.6152 and recall 0.7845. Decision tree gains recall but loses ROC-AUC, from 0.8545 to 0.8400. To keep the computational cost of nested tuning manageable, it is limited to two models: the two with the highest weighted ROC-AUC, CatBoost (0.9020) and logistic regression (0.8988). They also represent two different model families, gradient-boosted trees and a linear model.

Table 11: Class-weighted engineered models, mean over two folds.

Model	ROC-AUC	AP	F1	Fit s
CatBoost	0.9020	0.6976	0.6118	11.7
Logistic regression	0.8988	0.6873	0.6152	111.2
LinearSVC	0.8981	0.6855	0.6178	88.9
Random forest	0.8965	0.6850	0.6340	183.6
Decision tree	0.8400	0.5958	0.5527	45.9

Table 12: Effect of class weighting on the engineered representation (unweighted / weighted).

Model	Precision	Recall	BA
CatBoost	0.7320 / 0.4941	0.5244 / 0.8029	0.7430 / 0.8195
Logistic regression	0.7344 / 0.5060	0.5117 / 0.7845	0.7374 / 0.8159
LinearSVC	0.7440 / 0.5135	0.4924 / 0.7755	0.7293 / 0.8145
Random forest	0.7558 / 0.5892	0.4629 / 0.6861	0.7166 / 0.7954
Decision tree	0.6592 / 0.4364	0.4782 / 0.7536	0.7145 / 0.7798

15 Nested hyperparameter optimization

GridSearchCV optimizes ROC-AUC within two stratified inner folds and evaluates the selected pipeline on three outer folds. All preprocessing is included in the fitted pipeline. CatBoost searches $\{300, 600\}$ iterations, depths $\{5, 7\}$ and learning rates $\{0.03, 0.1\}$ (eight combinations); logistic regression searches $C \in \{0.1, 1, 10\}$ (three combinations). Both use class weighting and engineered features.

Table 13 reports the outer-fold means. CatBoost achieves ROC-AUC **0.9023**, AP **0.6986**, recall **0.8051** and balanced accuracy **0.8201**. Logistic regression has slightly higher F1 (0.6155 versus 0.6118) and precision (0.5061 versus 0.4934). CatBoost is selected for its stronger ranking performance and sensitivity, accepting more false positives. The complete nested experiment takes 37.14 minutes.

Table 13: Three outer-fold nested CV results with two-fold inner grid search.

Model	ROC-AUC	AP	F1	Precision	Recall	BA
CatBoost	0.9023 ± 0.0013	0.6986	0.6118	0.4934	0.8051	0.8201
Logistic regression	0.8992 ± 0.0012	0.6881	0.6155	0.5061	0.7851	0.8162

The ROC-AUC dispersion is the sample standard deviation across folds.

Optimization produces only a marginal increase in ROC-AUC relative to the preceding weighted experiment (0.9020 to 0.9023). This comparison is not like-for-like: the outer splits differ, and with three outer folds each model is trained on 67% of the data instead of 50% in two-fold screening. Part of this small gain may therefore come from the larger training set rather than from tuning.

The selected CatBoost parameters change from depth 6, 300 iterations and learning rate 0.1 to depth 7, 600 iterations and learning rate 0.03. Logistic regression changes from $C = 1$ to $C = 0.1$. Thus logistic regression benefits from stronger regularization, while CatBoost uses slower and more extensive boosting with a deeper tree.

Table 14 lists the selected parameters for each outer fold. All three outer folds select identical settings, which are fixed for the following ten-fold comparison. All selected values lie on the edge of the explored grid. The optimum may therefore lie outside the explored range, and a wider grid could yield further, probably small, gains.

Table 14: Selected parameters and ROC-AUC for each nested outer fold.

Model	Fold	Selected settings	Inner AUC	Outer AUC
CatBoost	1	depth 7, iterations 600, rate 0.03	0.9013	0.9032
CatBoost	2	depth 7, iterations 600, rate 0.03	0.9022	0.9008
CatBoost	3	depth 7, iterations 600, rate 0.03	0.9013	0.9028
Logistic regression	1	C = 0.1	0.8977	0.9001
Logistic regression	2	C = 0.1	0.8990	0.8979
Logistic regression	3	C = 0.1	0.8981	0.8995

16 Statistical comparison with the Wilcoxon test

The paired observations are the two models' ROC-AUC values on exactly the same ten stratified outer folds. Each respondent receives one out-of-fold prediction from each model. Table 15 summarizes these fixed-configuration runs and Figure 11 displays the foldwise scores.

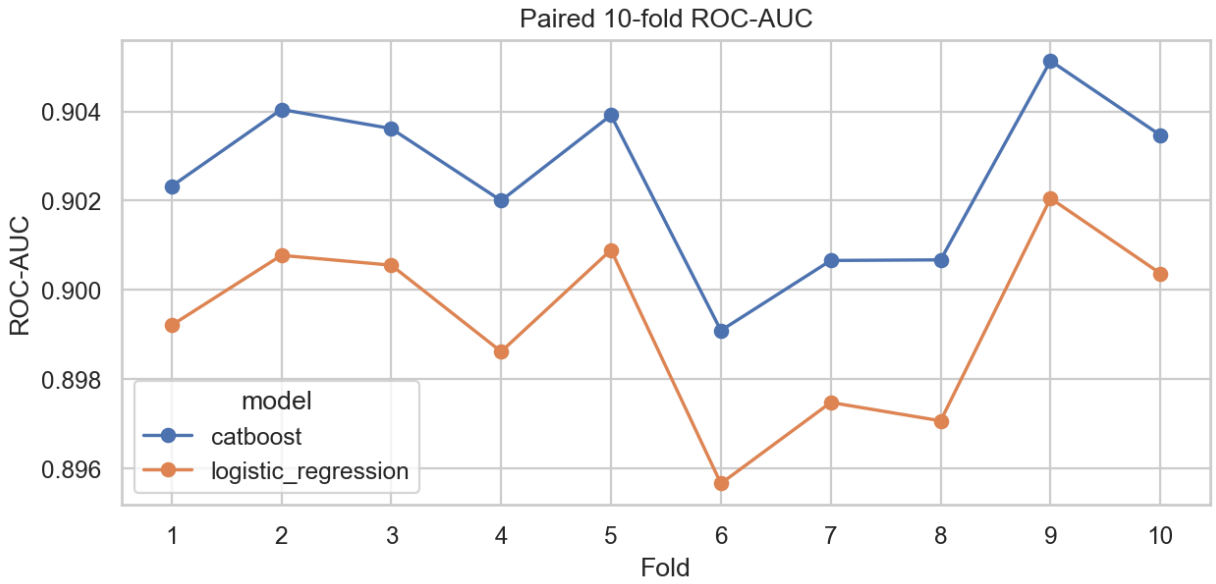


Figure 11: Paired ten-fold ROC-AUC for CatBoost and logistic regression.

CatBoost has higher ROC-AUC in all ten folds. Mean ROC-AUC is 0.90249 versus 0.89926 for logistic regression; the mean difference is **0.0032276** and the median difference is **0.0031516**. A two-sided Wilcoxon signed-rank test gives $W = 0$ and $p = 0.00195$, significant at the nominal 0.05 level. With ten pairs that all have the same sign, $p = 2/2^{10} \approx 0.00195$ is the smallest value the exact two-sided test can produce. The test therefore establishes that the direction of the difference is consistent, not that the effect is large: the gain is only about 0.003 AUC.

Table 15: Paired ten-fold evaluation of the fixed selected configurations.

Model	ROC-AUC	AP	F1	Precision	Recall	BA
CatBoost	0.9025 ± 0.0019	0.6989	0.6113	0.4921	0.8066	0.8203
Logistic regression	0.8993 ± 0.0020	0.6884	0.6153	0.5060	0.7849	0.8160

The ROC-AUC dispersion is the sample standard deviation across folds.

This test has two caveats: the training sets of the ten folds overlap, so the paired scores are not independent, and the hyperparameters were tuned on the same data. The result therefore shows a consistent advantage of CatBoost just on this dataset.

The aggregate out-of-fold confusion counts in Figure 12 explain the operating tradeoff. CatBoost detects 58,676 of 72,741 positive respondents and misses 14,065, with 60,569 false positives. Compared

with logistic regression, it detects 1,580 additional positive respondents while producing 4,816 additional false positives. This is consistent with prioritizing recall, but a deployment decision would require an explicit cost for missed cases and follow-up assessments.

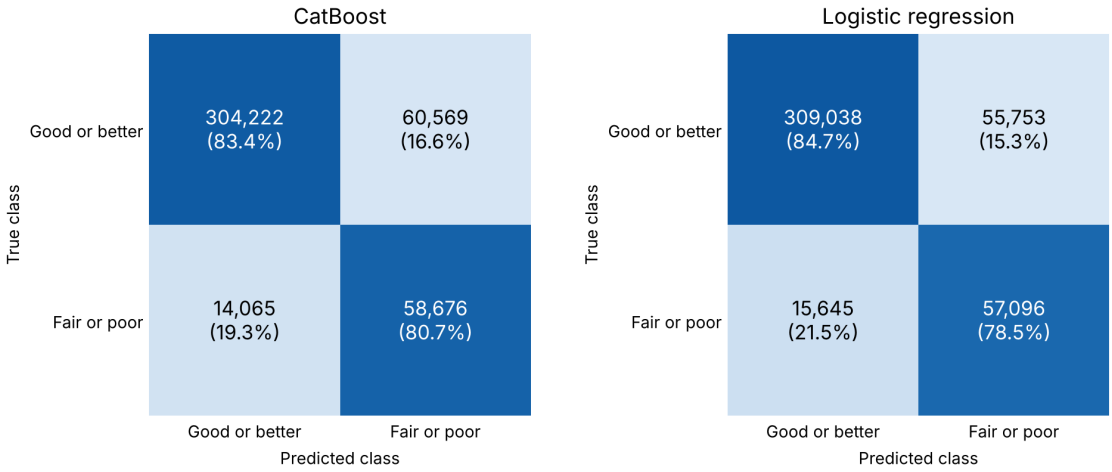


Figure 12: Aggregate ten-fold out-of-fold confusion matrices. Percentages are shares of each true class.

17 Error analysis

The three highest-confidence misclassifications are selected from CatBoost's ten-fold out-of-fold predictions, so the corresponding validation model did not train on the respondent being assessed. All three cases are **false positives**: the respondents rate their health as good or better while CatBoost assigns a very high score to fair or poor health. The tuned logistic regression misclassifies the same three respondents in the same folds, with even higher scores (Table 16), so these are errors made consistently by two different model families.

Table 16: Three highest-confidence CatBoost out-of-fold errors and the logistic-regression scores for the same respondents.

Row	Rating	Age	BMI	Physical	Mental	Conditions	Limits	CatBoost	LR
281993	Good	57	26.57	30	30	4	5	0.9953	0.9995
345486	Good	66	22.81	30	0	7	2	0.9936	0.9990
256508	Excellent	80	21.26	25	25	6	5	0.9931	0.9991

Rating is the original `GENHLTH` answer; the true class is 0 and both models predict class 1 for all rows. CatBoost and LR columns are out-of-fold scores for fair or poor health. Physical and mental columns are unhealthy days in the last 30 days; condition and limitation columns are affirmative-response counts.

Respondent 281993 rates their health as *good*. They are 57 years old, report 30 physically and 30 mentally unhealthy days and no physical activity, and confirm four conditions: stroke, COPD, diabetes and depression. Heart disease, both cancer questions and kidney disease are answered "don't know" and therefore do not count. They report five functional limitations (vision, cognition, walking, dressing and errands), have been out of work for more than a year and have elementary-level education.

Respondent 345486 also rates their health as *good*. They are 66 years old, report 30 physically unhealthy days but no mentally unhealthy days (code 88), smoke daily, are physically inactive and are unable to work. They confirm seven conditions: coronary heart disease, asthma, skin cancer, COPD, kidney disease, diabetes and arthritis, and two limitations (hearing and walking).

Respondent 256508 rates their health as *excellent*. Their recorded age is 80, which may mean 80 or older because of top-coding. They report 25 physically and 25 mentally unhealthy days, no

physical activity and inability to work, and confirm six conditions: heart attack, coronary heart disease, a non-skin cancer, depression, diabetes and arthritis, together with five functional limitations (vision, cognition, walking, dressing and errands).

We propose three complementary explanations. First, **the binarization boundary**: two of the three respondents chose "good", the category adjacent to "fair". The binary target collapses an ordinal scale between these two categories, so respondents with a heavy reported burden who still choose "good" are the hardest cases by construction; an ordinal or three-class formulation would treat them as near misses. Second, **subjective reference frames**: self-rated health is judged relative to the respondent's own expectations, age peers and adaptation to long-standing illness, none of which the questionnaire observes. Third, **inconsistent or erroneous answers**: an "excellent" rating combined with 25 unhealthy days, six diagnoses and five limitations, as for respondent 256508, is hard to reconcile and may reflect a misunderstanding of the scale or a recording error. All three respondents report only elementary-level education, which is compatible with comprehension difficulties, although three cases cannot establish this.

In all three cases the model's prediction is consistent with the detailed answers, and the error lies in the relation between those answers and the global rating. Forcing the model to fit such cases would largely mean fitting label noise.

18 Explainability with SHAP

18.1 Global interpretation

The selected CatBoost pipeline is refitted on all 437,532 respondents for interpretation. SHAP is computed on a random sample of **1,000 respondents**, with the first **200 transformed sample records** used as background. The beeswarm plot (Figure 13) orders features by mean absolute SHAP value and shows the direction and distribution of their contributions.

The model relies primarily on physical health, functional limitations, chronic disease burden, age-related comorbidity, BMI, diabetes and physical activity. Several highly ranked variables, such as `physical_days` and `_physical_status`, describe related aspects of physical health, so their importance is distributed across multiple encoded features.

Positive SHAP values move predictions toward fair or poor health, while negative values move them toward good or better health. Specifically, `physical_days_88.0` and `_physical_status_1.0` represent zero physically unhealthy days during the previous 30 days. Active indicators move predictions toward good or better health. In contrast, `_physical_status_3.0`, representing 14–30 unhealthy days, moves predictions toward fair or poor health (CDC, 2022b,c).

High values of `functional_limitations_count`, `comorbidity_count` and `age_x_comorbidity` mostly have positive SHAP values: multiple limitations and diseases, particularly at older recorded ages, raise the model output. Diagnosed diabetes (`diabetes_1.0`) and no physical activity (`exercise_2.0`) also shift predictions toward fair or poor health, whereas no diabetes (`diabetes_3.0`) shifts them in the opposite direction. Higher BMI generally raises the score, while categories corresponding to higher education and income tend to lower it.

18.2 Local explanations for selected errors

Figure 14 explains the high model output for respondent 256508; the waterfall plots of the other two respondents show the same pattern. The first two respondents report 30 physically unhealthy days and the third 25 days. Thus the 14–30-day indicator is active in all three, while the zero-day indicators are inactive, strongly shifting outputs toward fair or poor health. Additional positive contributions arise from high standardized counts of comorbidities and functional limitations, diabetes, low education and, for two respondents, being unable to work.

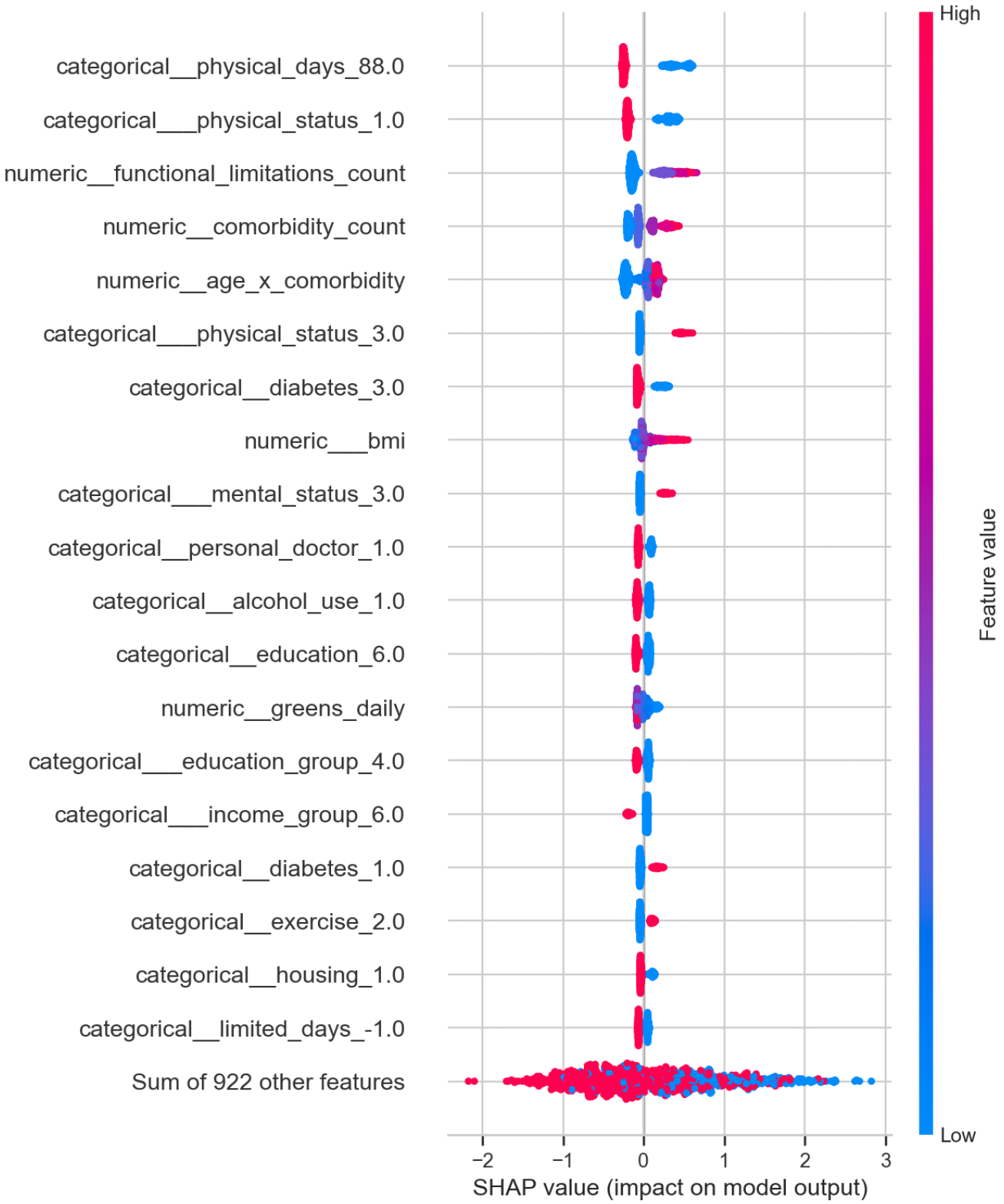


Figure 13: SHAP beeswarm showing the magnitude and direction of feature contributions. Red indicates higher encoded feature values and blue lower values.

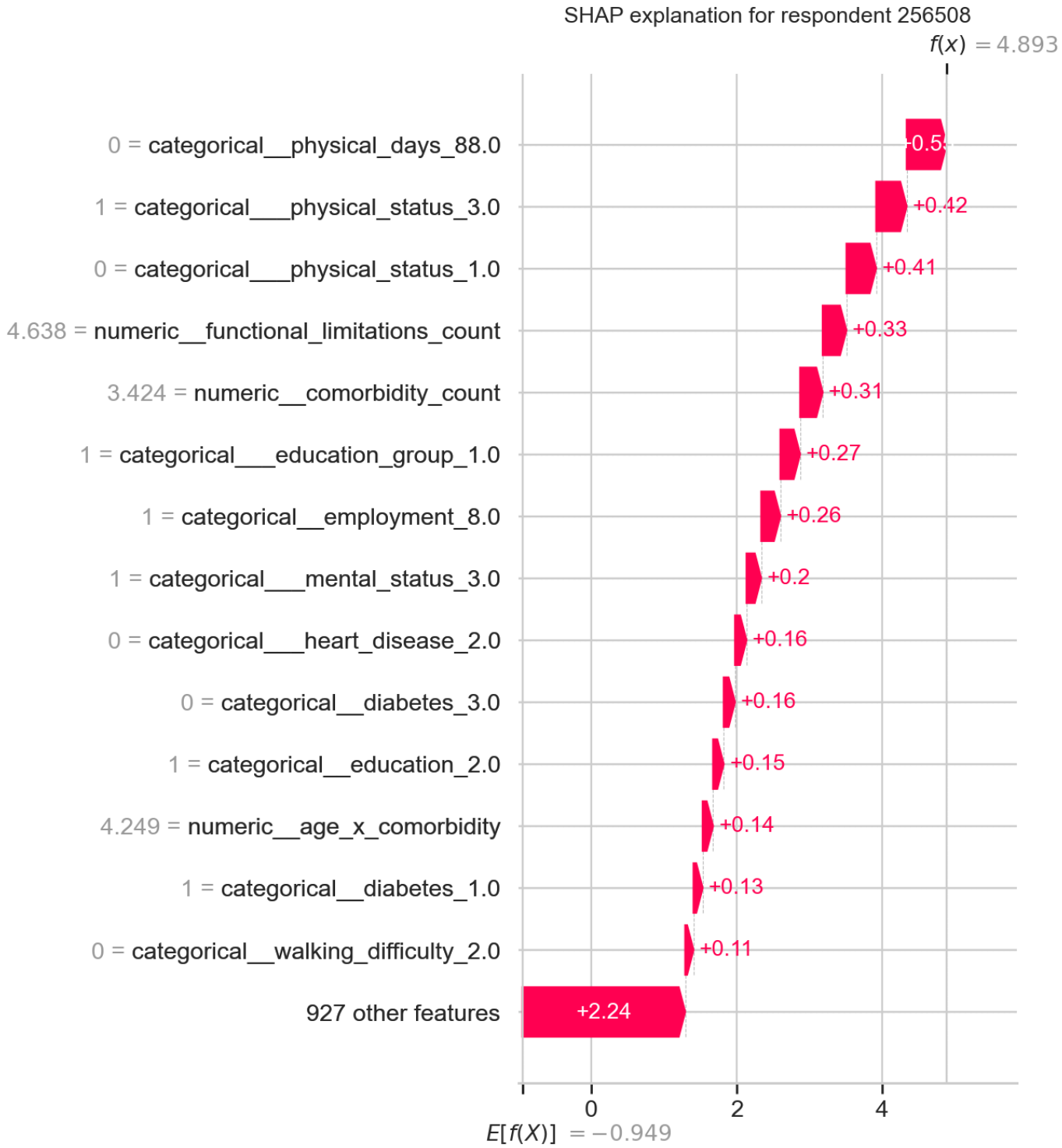


Figure 14: Local SHAP waterfall for respondent 256508 from the full-data refitted model. Red contributions increase the fair-or-poor output; blue contributions decrease it. The output is on the raw model scale.

19 Discussion, limitations and conclusions

Table 17 summarizes the decisions taken at each stage. The main result is consistent across feature experiments, nested tuning and the paired comparison: CatBoost gives the best ROC-AUC among the selected classifiers, while logistic regression remains close and offers slightly higher F1 and precision after tuning. All learned classifiers substantially outperform the dummy. CatBoost's nested ROC-AUC of 0.9023 and AP of 0.6986 show strong discrimination for concurrent self-rated health.

Table 17: Summary of experiments: best model at each stage and the resulting decision.

Stage	CV	Best model	ROC-AUC	AP	Recall	Decision
Dummy baseline	2-fold	–	0.5000	0.1663	0.0000	Reference level
Full features	2-fold	CatBoost	0.9018	0.6985	0.5246	Reference representation
SelectKBest (100)	2-fold	CatBoost	0.8956	0.6880	0.5085	Fastest; small loss in AUC
TruncatedSVD (100)	2-fold	LinearSVC	0.8958	0.6852	0.4777	Not retained; hurts tree models
RFECV (≈190)	2-fold	CatBoost	0.9004	0.6953	0.5226	Best compact alternative
Engineered	2-fold	CatBoost	0.9019	0.6988	0.5244	Equivalent to full; retained for interpretability
Class weighting	2-fold	CatBoost	0.9020	0.6976	0.8029	Adopted: recall +0.28 at equal AUC
Nested tuning	3×2	CatBoost	0.9023	0.6986	0.8051	Depth 7, 600 iterations, rate 0.03
Paired comparison	10-fold	CatBoost	0.9025	0.6989	0.8066	Beats LR in 10/10 folds ($p = 0.00195$); **final model**

Both feature-selection hypotheses are supported: SelectKBest keeps most of the discrimination with 100 features and the largest runtime reduction, and RFECV retains near-full CatBoost performance with about 190 features during outer evaluation. SVD captures substantial variance but does not improve discrimination. The feature-engineering hypothesis is not supported as a material performance gain: the CatBoost improvement is only 0.000056 AUC. Nevertheless, disease and limitation counts aid interpretation. Class weighting markedly increases recall at the cost of precision which is a tradeoff.

Several limitations define the scope of these conclusions. First, self-reports, structural blanks and uncertainty codes complicate interpretation; affirmative-response counts can understate burden when answers are unknown. Median imputation for EDA also affects displayed numerical distributions. Second, the target and the strongest symptom and limitation predictors are measured in the same questionnaire. Removing `_RFHLTH` prevents direct leakage, but retaining unhealthy-day counts means that the analysis partly measures concurrent construct overlap (Section 3).

Within this scope, **class-weighted CatBoost with engineered features is the selected model**, using 600 iterations, depth 7 and learning rate 0.03. Logistic regression is a strong alternative when precision, a simpler model or implementation constraints matter more than a small AUC gain.

Ethics and Use of Generative AI

A generative AI tool, Claude, was used as a support tool at four points of this project: (1) to help map dataset columns to their definitions and response codes in the CDC codebook and calculated-variable documentation; (2) to help write and adjust the plotting code for figures; (3) to help format the tables of this report in LaTeX; and (4) to polish the language of this report. All other parts of the work, including the problem framing, data cleaning decisions, experimental design, implementation and execution of the experiments, and the interpretation of the results, were carried out by the author. All AI suggestions were reviewed, checked against the documentation and the obtained results, and adapted before being incorporated. No assistance from other people was received. I assume responsibility for and authorship of this work.

A Supplementary material

Table 18: General health ratings after excluding invalid or missing target responses.

Code	Rating	Respondents	Percent
1	Excellent	77,741	17.77
2	Very good	149,112	34.08
3	Good	137,938	31.53
4	Fair	54,736	12.51
5	Poor	18,005	4.12
Class 0 Good or better		364,791	83.37
Class 1 Fair or poor		72,741	16.63

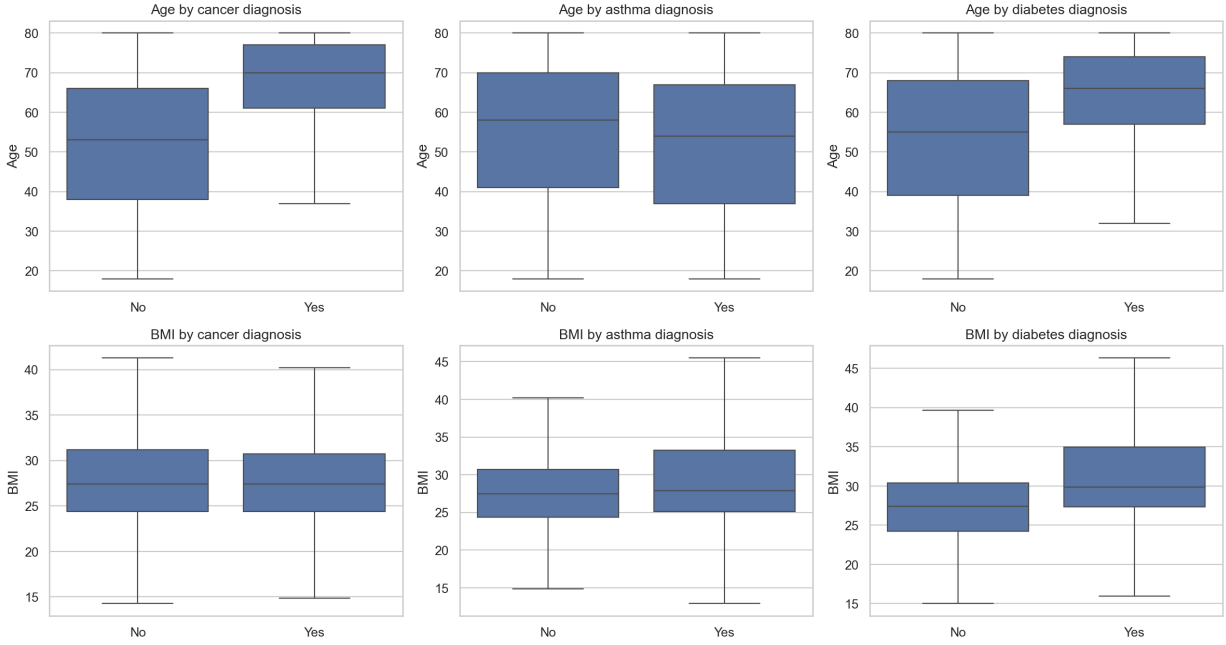


Figure 15: Age and BMI by affirmative diagnosis indicator for cancer, asthma and diabetes. "No denotes all responses not coded as an affirmative diagnosis.

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